

Raghav Senthil Kumar

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, Agentic AI, BuildIllinois, Innovation LLC

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Coded and deployed the cloud-hosted document search engine behind a competitive high school debate-research SaaS platform in *Python (FastAPI)* on *AWS*, working in an *Agile* team of **10 engineers** for **10K+ site visitors**
- Built a data pipeline that indexed **600K+ documents** on *Elasticsearch*, returning results in **under 1 second**
- Improved data quality with a de-duplication algorithm that caught **30%** of evidence documents as identical
- Wrote a text-retrieval algorithm that validates the source URLs and filters out statistical outliers with **IQR**

Product Development Intern | Jetson

January 2025 – February 2025

- Contributed to the execution of key tasks via **7 iterations of UI/UX testing** on **3 experimental features**
- Provided input on project strategies in a **5-page report** to optimize user engagement, delivered to Jetson's CEO
- Condensed user onboarding to **5 pages** from **12** in *Figma*, then **A/B tested** the new flow on 30 test users

Data Analytics Intern | STEM E

June 2024 – July 2024

- Investigated the **3 years** of TikTok data spanning **200+ posts** using *Python (pandas)*, *Power BI* and *Excel*
- Surfaced a clear pattern: “science fun facts” posts were **7%** of all content yet still drove **69.3%** of total views
- Shared the insight as a “fun facts” content strategy; the Executive Director adopted it and engagement rose **24%**

PROJECTS

Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Directed a team of **4** on **Kamion (YC S22)**'s pricing algorithm, winning **2nd at UIUC Founder's Hackathon**
- Built a **15,000-image** truck-listing dataset through scraping, vetting it with de-duplication and non-USD filtering
- Prompted a *Claude Sonnet 4.6* agent via a *JSON schema* to read condition, brand, model and year from photos
- Debugged the schema against wrong predictions until **10 consecutive images** were correct on **3 features**
- Priced trucks with *quantile regression* on the agent's extracted features, hitting **87% accuracy** on **30 samples**

Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Deployed a **RAG** AI agent on *Claude Sonnet 4.5* that retrieves session-scoped documents from a vector store
- Automated paper ingestion via *OpenAlex*, backfilling gaps with *Semantic Scholar* metadata in a dual-API pipeline
- Loaded papers into a *Supabase (PostgreSQL)* table and wrote custom *SQL* filters by field, year and citation count
- Developed an interactive *React Flow* graph view of outbound citations for papers across **26 research fields**

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Ran **4 Random Forest** machine learning models in a *Python FastAPI* back end on live data ingested from a school's **3 tools** to detect anomalies and deliver actionable steps via a *Gemma* Voice LLM on a *Next.js* dashboard
- Prototyped and built **3 edge AI tools** for data collection on a school's energy, water and food consumption
- Steered a team of **5** from ideation to presentation, earning a **USAII Global Hackathon Finalist** placement

EcoBin | [Research Paper](#) | [Source Code](#)

March 2026 – May 2026

- Achieved **96.13% accuracy** classifying waste items into 4 disposal pathways with a two-stage *EfficientNetV2-S* deep learning network (**4.39M** parameters) trained in *PyTorch* on **15K+ images**, then published a paper on *arXiv*
- Produced **9K** contamination-detection images via *U2-Net* segmentation and alpha-compositing in a pipeline
- Placed **2nd at the Arizona State Science Fair** and posted **2 original datasets** on *Kaggle* for the project

SKILLS

Programming Languages: Python, Java, C++, SQL, JavaScript, HTML/CSS

Frameworks/Platforms: FastAPI, Flask, PostgreSQL, Supabase, Elasticsearch, AWS, Google Cloud, Next.js

AI/ML: Claude API, PyTorch, TensorFlow, scikit-learn, Hugging Face, Keras, OpenCV, Gemma 4, Suprememory

Data & Analytics: pandas, NumPy, Power BI, Microsoft Excel, A/B Testing, Hootsuite

Concepts/Practices: Artificial Intelligence, Coding, Data Collection, Data Engineering, Code Testing